Mateusz Rajzer

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github.com/mateuszrajzermath-dot/Selberg-CLT-Analysis

Profile

Self-directed Master's student specializing in Arithmetic Geometry and Number Theory. Although formally enrolled in Financial Mathematics, my independent research is focused on pure mathematics—particularly modular forms, the Sato-Tate conjecture, and computational number theory. My recent participation in the Arizona Winter School 2026 directly inspired my thesis on the Fourier coefficients of newforms. I aim to pursue a PhD to continue developing these directions within a formal, rigorous research environment.

Education

University of Rzeszow, Rzeszow, Poland
Master of Science in Mathematics **Expected 06/2026**
Specialization: Financial Mathematics
Master's Thesis: *Analysis of the Distribution of Normalized Fourier Coefficients for a Fixed Newform in the Context of the Sato-Tate Conjecture*
Supervisor: Prof. Oleh Lopuszanski

Rzeszow University of Technology, Rzeszow, Poland
Bachelor of Science in Mathematics **2023**

Advanced Academic Programs

- **Arizona Winter School 2026: Computational Aspects of Arithmetic Geometry**
  Southwestern Center for Arithmetic Geometry, Tucson, AZ, USA **03/2026**
  Selected participant for the intensive winter school focusing on explicit computations of modular forms and arithmetic geometry.

Research Interests

- **Arithmetic Geometry & Number Theory:** Modular Forms, Newforms, Sato-Tate Conjecture, L-Functions, Galois Representations, Selberg's Central Limit Theorem.
- **Computational Methods:** Explicit Computations in Arithmetic Geometry, Empirical Analysis in Number Theory (SageMath, Python).
- **Operator Algebras & Non-commutative Geometry:** Bost-Connes Model, C*-Algebras, KMS States.

Conference Talks & Presentations

- **"(Upcoming) Manin Symbols and Computations in Spaces of Weight 2 Modular Forms"** (Talk)
  National Session of Mathematics Students (Ogólnopolska Sesja Studentów Matematyki), Kraków, Poland **04/2026**

- **“From Arithmetic Chaos to Algebraic Structure: An Introduction to Iwasawa Theory”** (Talk)
  To Pikuś Conference (University-level), University of Rzeszow, Poland **12/2025**
- **“The Taniyama-Shimura-Weil Theorem and Elliptic Curves”** (Talk)
  Omatko Conference (National), University of Wrocław, Poland **12/2025**
- **“Empirical Analysis of Selberg’s Central Limit Theorem for the Riemann Zeta Function”** (Talk)
  Elements Conference (National), AGH University, Kraków, Poland **10/2025**
- **“The Riemann Hypothesis and Analytic L-Functions”** (Talk)
  V Conference: Faces of Mathematics (National), University of Rzeszow, Poland **06/2025**
- **“An Introduction to Freudenthal’s Suspension Theorem in Homotopy Theory”** (Talk)
  Sempowisko (International Student Conference), Jagiellonian University, Kraków, Poland **05/2025**
- **“On the Riemann Hypothesis and the Nature of L-Functions”** (Talk)
  To Pikuś Conference (University-level), University of Rzeszow, Poland **05/2025**
- **“An Introduction to Modern Galois Theory and Field Extensions”** (Talk)
  Numbers, People, Space: Interdisciplinary Approach. (International Conference), University of Rzeszow, Poland **04/2025**
- **“C*-Algebras and their Applications in Operator Theory”** (Talk)
  To Pikuś Conference (University-level), University of Rzeszow, Poland **12/2024**
- **“An Introduction to C*-Algebras in Quantum Mechanics”** (Poster)
  Omatko Conference (National), University of Wrocław, Poland **12/2024**
- **“A Brief History of Diophantine Equations”** (Talk)
  Mathematics School (Regional), Tarnów, Poland **05/2022**

Teaching & Outreach Experience

- **Mathematics Science Club Activity**
  University of Rzeszow **2024 – Present**
  Active member and organizer of mathematical discussions and student seminars, fostering a collaborative academic environment.
- **Mathematics Instructor**
  Conducted extracurricular mathematics classes for 8th-grade students, breaking down complex concepts and developing their analytical problem-solving skills.

Honors & Awards

- **Witold Wilkosz Award (III Degree)**
  Awarded by the Polish Mathematical Society (PTM) for the outstanding student work popularizing mathematics: "Dziennik Dociekań o Smokach Czyli Równania na Smoczych Skrzydłach". **2025**
- **Rector’s Scholarship for Academic Excellence**
  Awarded for outstanding academic achievements in the mathematics department for the 2025/2026 academic year.
- **Award for Best Student Presentation (3rd Place)**
  Elements Conference (National, AGH University) for the talk on Selberg’s Central Limit Theorem. **10/2025**

Technical Skills

- **Programming:** SageMath, Python (NumPy, SciPy)

- **Languages:** Polish (Native), English (Fluent)
- **Tools:** LaTeX